

Lab 5 PHYS 5394 - Ben Carlson

#1. Using the definition of expectation in terms of the pdf,

$$E[f(X)] = \int_{-\infty}^{\infty} f(x) p_x(x) dx$$

show that

(a) $E[f(W) + g(U)] = E[f(W)] + E[g(U)]$ for any two random variables U and W

(b) $E[X + a] = E[X] + a$ where a is a constant while X is a random variable

(c) $E[aX] = aE[X]$

(d) $var[aX] = a^2 var[X]$ where $var[]$ means variance

a)

$$\begin{aligned} E[f(W) + g(U)] &= \int_{-\infty}^{\infty} (f(W) + g(U)) p_x(x) dx \\ &= \int_{-\infty}^{\infty} f(W) p_x(x) dx + \int_{-\infty}^{\infty} g(U) p_x(x) dx \\ &= E[f(W)] + E[g(U)] \end{aligned}$$

b)

$$\begin{aligned} E[X + a] &= \int_{-\infty}^{\infty} (X + a) p_x(x) dx \\ &= \int_{-\infty}^{\infty} X p_x(x) dx + a \int_{-\infty}^{\infty} p_x(x) dx \\ &= E[X] + a \cdot 1 \end{aligned}$$

c)

$$\begin{aligned} E[aX] &= \int_{-\infty}^{\infty} aX p_x(x) dx \\ &= a \int_{-\infty}^{\infty} X p_x(x) dx \\ &= aE[X] \end{aligned}$$

d)

$$\begin{aligned}
 \text{var}[aX] &= E[(aX - E[aX])^2] \\
 &= \int_{-\infty}^{\infty} (aX - E[aX])^2 p_x(x) dx \\
 &= \int_{-\infty}^{\infty} (aX - aE[X])^2 p_x(x) dx \\
 &= a^2 \int_{-\infty}^{\infty} (X - E[X])^2 p_x(x) dx \\
 &= a^2 \text{var}[X]
 \end{aligned}$$

#2. Use the linearity property of expectation to show that the variance of the sum of two random variables $Z = X + Y$ is given by $\text{var}[Z] = \text{var}[X] + \text{var}[Y] + 2C_{XY}$. Hint: $\text{var}[Z] = E[(Z - E[Z])^2] = E[(X + Y - E[X + Y])^2]$.

$$\begin{aligned}
 \text{var}[Z] &= E[(Z - E[Z])^2] \\
 &= E[(X + Y - E[X + Y])^2] \\
 &= \int_{-\infty}^{\infty} (X + Y - E[X + Y])^2 p_x(x) dx \\
 &= \int_{-\infty}^{\infty} (X^2 + 2XY + Y^2 - 2(X + Y)E[X + Y] + E[X + Y]^2) p_x(x) dx \\
 &= \int_{-\infty}^{\infty} (X^2 + 2XY + Y^2 - 2(X + Y)(E[X] + E[Y]) + E[X]^2 + 2E[X]E[Y] + E[Y]^2) p_x(x) dx \\
 &= \int_{-\infty}^{\infty} (X^2 - 2XE[X] + E[X]^2 + Y^2 - 2YE[Y] + E[Y]^2 + 2XY + 2E[X]E[Y] - 2(YE[X] + XE[Y])) p_x(x) dx \\
 &= \int_{-\infty}^{\infty} ((X - E[X])^2 + (Y - E[Y])^2 + 2(XY + E[X]E[Y] - YE[X] - XE[Y])) p_x(x) dx \\
 &= \text{var}[X] + \text{var}[Y] + 2 \int_{-\infty}^{\infty} (XY + E[X]E[Y] - YE[X] - XE[Y]) p_x(x) dx \\
 &= \text{var}[X] + \text{var}[Y] + 2C_{XY}
 \end{aligned}$$